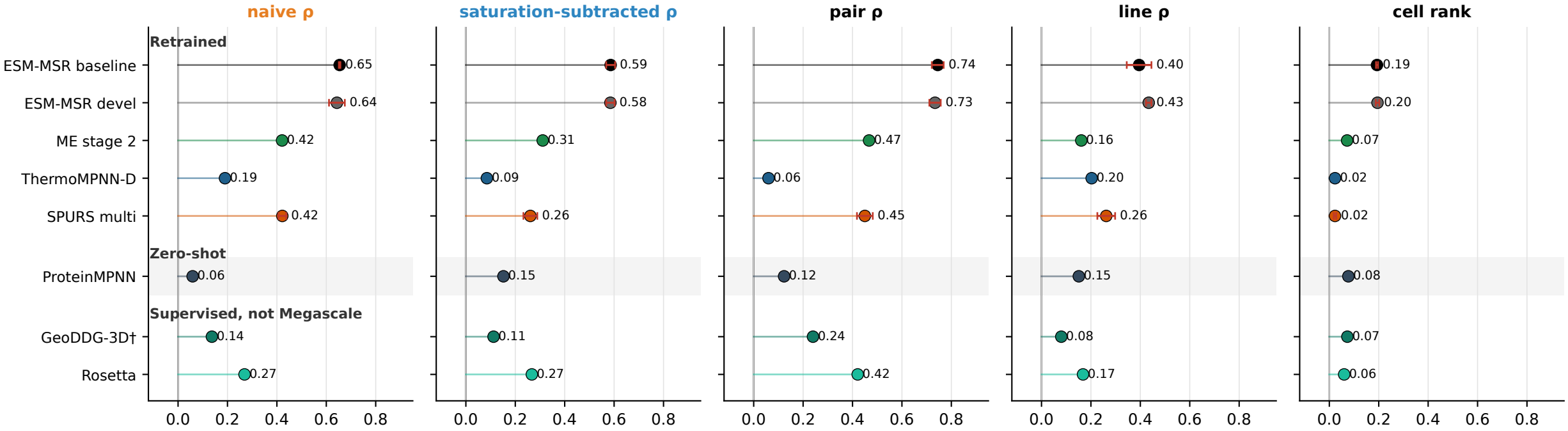
Validation: epistasis predicted level by level, 8,494 doubles with both singles measured, 24 position pairs



† trained on S8754 (ProThermDB + ThermoMutDB), not on Megascale. ESM3-small-open and ProSST are not in the epistasis tables; ESM3ΔG is left out (overlap with its MGnify fine-tune).
Error bars: sd over 3 seeds where the table has seeds (ESM-MSR rows, SPURS multi); other rows are single checkpoints or public models. Numbers are the benchmark document's (tables 2.1 / 2.2).